1 Families of Complexes

1.1 First Family

Let X be a smooth projective variety of dimension n, and D an ample divisor. The adjoint bundle $\omega_X(D)$ induces a map

$$X \dashrightarrow |K_X + D|^\vee = \mathbb{P}(H^0(X, \omega_X(D))^\vee).$$

This ambient projective space can be considered a family of Bridgeland stable objects. Given an individual closed point $\langle f \rangle \in |K_X + D|^\vee$, we can identify

$$\begin{aligned} |K_X + D|^\vee &\cong \mathbb{P}(H^0(X, \omega_X(D))^\vee) \\ &\cong \mathbb{P}H^n(X, \mathcal{O}_X(-D)) \\ &\cong \mathbb{P}\mathrm{Hom}_{D^b(X)}(\mathcal{O}_X, \mathcal{O}_X(-D)[n]), \end{aligned}$$

to consider f as a morphism $\mathcal{O}_X \to \mathcal{O}_X(-D)[n]$, and taking the cone (our convention is to put cones in front), we get distinguished triangle

$$\to C_f \to \mathcal{O}_X \to \mathcal{O}_X(-D)[n] \to,$$

where C_f only depends on f up to scaling, so we can really identify $\langle f \rangle$ with C_f.

Now, to do this in families, we will build a complex $C_{|K_X+D|^\vee}$ on $X \times |K_X+D|^\vee$ (i.e., a family over $|K_X + D|^\vee$ of complexes on X). To build this complex, we write down a distinguished triangle

$$\to C_{|K_X+D|^\vee} \to \mathcal{O}_{X\times|K_X+D|^\vee} \to \mathcal{O}_X(-D) \boxtimes \mathcal{O}_{|K_X+D|^\vee}(1)[n] \to$$

which is determined by (the k-span of) a morphism in

$$\mathrm{Hom}_{D^b(X\times|K_X+D|^\vee)}(\mathcal{O}_{X\times|K_X+D|^\vee}, \mathcal{O}_X(-D) \boxtimes \mathcal{O}_{|K_X+D|^\vee}(1)[n]),$$

which gets identified with

$$\begin{aligned} H^n(X \times |K_X + D|^\vee, \mathcal{O}_X(-D) \boxtimes \mathcal{O}_{|K_X+D|^\vee}(1)) &\cong H^n(X, \mathcal{O}_X(-D)) \otimes_k H^0(|K_X + D|^\vee, \mathcal{O}_{|K_X+D|^\vee}(1)) \\ &\cong H^0(X, \omega_X(D))^\vee \otimes_k H^0(X, \omega_X(D)) \\ &\cong \mathrm{End}_k(H^0(X, \omega_X(D))) \qquad (1) \end{aligned}$$

which has a natural element, being the identity map. We will denote this morphism ε. Now, we should see that the cone over this map, which we denote $C_{|K_X+D|^\vee} := C_\varepsilon$, actually assembles the various C_f into a single family on $|K_X + D|^\vee$. That is, we claim the following.

Proposition 1.1. Let $\langle f \rangle \in |K_X + D|^\vee$ be a closed point, and

$$i : X \cong X \times \{\langle f \rangle\} \hookrightarrow X \times |K_X + D|^\vee$$

be the inclusion of the fiber over $\langle f \rangle$. Then,

$$Li^* C_{|K_X+D|^\vee} \cong C_f.$$

Proof. Before we start, we fix notation. Abbreviate $\mathbb{P} := |K_X + D|^\vee$. Pick a basis $e_1, \ldots, e_\ell$ for $H^0(X, \omega_X(D))^\vee$, with dual basis $x_1, \ldots, x_r$. Write $f = \sum a_i e_i$.

Now, to study the pullback $Li^* C_{\mathbb{P}}$, we study the triangle that defines it. The functor Li^* is exact, so when we apply it to the triangle

$$C_{\mathbb{P}} \to \mathcal{O}_{X \times \mathbb{P}} \xrightarrow{\varepsilon} \mathcal{O}_X(-D) \boxtimes \mathcal{O}_{\mathbb{P}}(1)[n] \to C_{\mathbb{P}}[1],$$

we will again get an exact triangle. We need to understand this 3rd term of the triangle. Pulling back, we get

$$Li^*(\mathcal{O}_X(-D) \boxtimes \mathcal{O}_{\mathbb{P}}(1)) \cong i^* p^* \mathcal{O}_X(-D) \otimes i^* q^* \mathcal{O}_{\mathbb{P}}(1),$$

where $i^* p^* \mathcal{O}_X(-D) \cong \mathcal{O}_X(-D)$, and $i^* q^* \mathcal{O}_{\mathbb{P}}(1)$ can be computed by pulling back instead along the equal composition

$$X \cong X \times \{\langle f \rangle\} \to \{\langle f \rangle\} \xhookrightarrow{j} \mathbb{P},$$

where now the pullback $j^* \mathcal{O}_{\mathbb{P}}(1)$ is clearly the trivial line bundle $\mathcal{O}_{\{\langle f \rangle\}}$, but we need to fix this isomorphism up to scaling by requiring that $j^* x_i = a_i \in k = \mathcal{O}_{\{\langle f \rangle\}}$ on the nose. So in total, the pullback of the 3rd term is $\mathcal{O}_X(-D)$.

This means our pulled back exact triangle is of the form

$$Li^* C_{\mathbb{P}} \to \mathcal{O}_X \xrightarrow{Li^* \varepsilon} \mathcal{O}_X(-D)[n] \to Li^* C_{\mathbb{P}}[1],$$

so our task is to verify that $Li^* \varepsilon$ agrees with f (up to scaling). So, this amounts to computing Li^* on morphisms

$$\mathrm{Hom}_{X \times \mathbb{P}}(\mathcal{O}_{X \times \mathbb{P}}, \mathcal{O}_X(-D) \boxtimes \mathcal{O}_{\mathbb{P}}(1)[n]) \to \mathrm{Hom}_X(\mathcal{O}_X, \mathcal{O}_X(-D)[n]).$$

Essentially by the triangle/zigzag axiom of the adjunction $Li^* \dashv Ri_*$, this can be computed by first postcomposing with the unit morphism $1 \Rightarrow Ri_* Li^*$, and then applying the adjunction isomorphism, as in

$$\begin{aligned} \mathrm{Hom}_{X \times \mathbb{P}}(\mathcal{O}_{X \times \mathbb{P}}, \mathcal{O}_X(-D) \boxtimes \mathcal{O}_{\mathbb{P}}(1)[n]) &\to \mathrm{Hom}_{X \times \mathbb{P}}(\mathcal{O}_X, Ri_* Li^*(\mathcal{O}_X(-D) \boxtimes \mathcal{O}_{\mathbb{P}}(1))[n]) \\ &\cong \mathrm{Hom}_{X \times \mathbb{P}}(\mathcal{O}_{X \times \mathbb{P}}, i_* \mathcal{O}_X(-D)) \\ &\xrightarrow{\cong} \mathrm{Hom}_X(i^* \mathcal{O}_{X \times \mathbb{P}}, \mathcal{O}_X(-D)), \end{aligned}$$

which, after translating to cohomology, becomes the usual pullback map

$$H^n(X \times \mathbb{P}, \mathcal{O}_X(-D) \boxtimes \mathcal{O}_{\mathbb{P}}(1)) \to H^n(X, \mathcal{O}_X(-D))$$

along i.

Now, we chase through everything. Recall our basis $e_1, \ldots, e_\ell$ of $H^n(X, \mathcal{O}_X(-D))$ and our dual basis, under the Serre duality pairing, $x_1, \ldots, x_\ell$ for $H^0(X, \omega_X(D))$.

Our first series of identifications takes $\varepsilon = \mathrm{id} \in \mathrm{End}_k(H^0(X, \omega_X(D)))$, and maps

$$\begin{aligned} \mathrm{End}_k(H^0(X, \omega_X(D))) &\longrightarrow H^n(X, \mathcal{O}_X(-D)) \otimes_k H^0(X, \omega_X(D)) \\ \varepsilon &\longmapsto \textstyle\sum_i e_i \otimes x_i. \end{aligned}$$

We then apply the Kunneth formula and restriction to get

$$\begin{aligned} H^n(X, \mathcal{O}_X(-D)) \otimes_k H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(1)) &\longrightarrow H^n(X \times \mathbb{P}, \mathcal{O}_X(-D) \boxtimes \mathcal{O}_{\mathbb{P}}(1)) \longrightarrow H^n(X, \mathcal{O}_X(-D)) \\ e_i, x_i &\longmapsto e_i \otimes x_i \longmapsto e_i x_i|_{\langle f \rangle} = j^* x_i \cdot e_i, \end{aligned}$$

where $j : \{\langle f \rangle\} \hookrightarrow \mathbb{P}$ is from before. Recall that we fixed our identifications so that $j^* x_i = a_i$, so indeed we see that the image of $\sum_i e_i \otimes x_i$ is f. Thus, $Li^* C_{\mathbb{P}} \cong C_f$. □

2 Second Family

Assume D is sufficiently positive so that $K_X + D$ is very ample, so $e : X \to |K_X + D|^\vee$ is an embedding. When we move our stability condition across a wall, we will destabilize exactly the complexes corresponding to $x \in X \hookrightarrow |K_X + D|^\vee$. It turns out that the correct thing to do is to replace $X \subseteq |K_X + D|^\vee$ with the projectivization of its normal bundle (i.e., a blowup) and to show that

$$Y = \mathrm{Bl}_X |K_X + D|^\vee$$

supports a new family of complexes on X, which will then be stable across the wall.

2.1 Complexes on X from X

Before we build a family of complexes on Y, we will focus on understanding the unstable locus $e : X \hookrightarrow |K_X + D|^\vee$. First, we will get an alternative distinguished triangle describing a complex C_{e_x} over a single point of $x \in X$. Next, we will understand globally the family of complexes on the unstable locus, by getting an alternative distinguished triangle describing the complex $C_X := Le^* C_{|K_X + D|^\vee}$.

To find these triangles, we will appeal to the octahedral axiom.

Remark 2.1. Recall the analogy between the third isomorphism theorm and the octahedral axiom: if we denote the cone (for a moment placing them on the right) over a morphism $A \to C$ by C/A, so that C/A fits in the distinguished triangle

$$\to C \to C/A \to A[1] \to,$$

and if we have a factorization $A \to B \to C$ (not a distinguished triangle) where each morphism has respective cone B/A, C/B, our notation suggests a distinguished triangle

$$\to B/A \to C/A \to C/B \to .$$

This existence of this triangle, together with several commutative diagrams, is the content of the octahedral axiom. For the purposes of destabilizing, this can be useful if for example B/A has greater slope than C.

2.1.1 Unstable complexes (pointwise)

First, we understand the complexes corresponding to $x \in X$, i.e., the C_{e_x}. We will fit these into distinguished triangles, which will eventually give the walls that destabilize (when we have the right numerical conditions).

Recall that a closed point $x \in X$ (up to scaling) gives an element

$$e_x \in H^0(X, \omega_X(D))^\vee \cong H^n(X, \mathcal{O}_X(-D)[n])$$

which is evaluating sections at x, which corresponds to (up to scaling) a morphism in the derived category $e_x : \mathcal{O}_X \to \mathcal{O}_X(-D)[n]$. We claim that we can factor this as a composition

$$\mathcal{O}_X \xrightarrow{\rho} \mathcal{O}_x \to \mathcal{O}_X(-D)[n],$$

which is to say, that pullback by ρ

$$\mathrm{Hom}(\mathcal{O}_X, \mathcal{O}_X(-D)[n]) \xleftarrow{\rho^*} \mathrm{Hom}(\mathcal{O}_x, \mathcal{O}_X(-D)[n])$$

has e_x in its image. By Serre duality, ρ^* corresponds to a morphism

$$\begin{gathered}\mathrm{Hom}(\mathcal{O}_X(-D), \omega_X)^\vee \leftarrow \mathrm{Hom}(\mathcal{O}_X(-D), \mathcal{O}_x \otimes \omega_X)^\vee, \\ H^0(X, \omega_X(-D))^\vee \leftarrow H^0(\{x\}, \omega_X(-D)|_x)^\vee\end{gathered}$$

which is dual to the map $\omega_X(-D) \to \omega_X(-D)|_x$. In other words, the dual of this map "is" the inclusion $\{x\} \hookrightarrow |K_X + D|^\vee$. Finally, asking for our factorization now amounts to asking that e_x (in the left hand side), which is evaluation at x (up to scaling), comes from restricting $\omega_X(-D)$ to x, which is tautological. So, indeed, we get our factorization $\mathcal{O}_X \to \mathcal{O}_x \to \mathcal{O}_X(-D)[n]$.

Now, we denote this second morphism by $\mathcal{O}_x \xrightarrow{x} \mathcal{O}_X(-D)[n]$, and we apply the octahedral axiom to our factorization to get the triangle

$$\to C_\rho \to C_{e_x} \to C_x \to,$$

and C_ρ is just $\mathcal{I}_x$ by the usual ideal sheaf exact sequence, so our triangle simplifies to

$$\to \mathcal{I}_x \to C_{e_x} \to C_x \to .$$

So, with our goal of understanding C_{e_x}, we try to understand C_x better. For this purpose, we try to understand the triangle (**??**) defining C_x by applying derived dual $(-)^\vee = R\mathcal{H}om(-, \mathcal{O}_X)$. Applying the functor term-by-term, first we apply Grothendieck duality along $i : \{x\} \to X$ (which has relative dimension $-n$) to get

$$R\mathcal{H}om_X(Ri_*\mathcal{O}_x, \mathcal{O}_X) \cong Ri_* R\mathcal{H}om_x(\mathcal{O}_x, i^*\omega_X^{-1} \otimes i^*\mathcal{O}_X[-n]) \cong \mathcal{O}_x[-n].$$

We also have

$$R\mathcal{H}om(\mathcal{O}_X(-D)[n], \mathcal{O}_X) \cong \mathcal{O}_X(D)[-n],$$

which together give us the triangle

$$\leftarrow C_x^\vee \leftarrow \mathcal{O}_x[-n] \leftarrow \mathcal{O}_X(D)[-n] \leftarrow$$

which identifies

$$C_x^\vee \cong \mathcal{I}_x(D)[1-n],$$

so

$$C_x \cong \mathcal{I}_x^\vee(-D)[n-1].$$

Finally, this means that C_{e_x} fits into the triangle

$$\to \mathcal{I}_x \to C_{e_x} \to \mathcal{I}_x^\vee(-D)[n-1] \to .$$

2.1.2 Unstable complexes (globally)

Even better than a triangle describing a single C_{e_x}, we will build an exact triangle (on $X \times X$) describing the assembled C_{e_x} as a family C_X over X.

The family C_X which is given by restricting the triangle defining $C_{|K_X+D|^\vee}$, i.e.,

$$\to C_X \to \mathcal{O}_{X\times X} \xrightarrow{\varepsilon} \mathcal{O}_X(-D) \boxtimes \omega_X(D)[n] \to .$$

Now, to repeat the story, we need a factorization of ε

$$\mathcal{O}_{X\times X} \xrightarrow{\rho} \mathcal{O}_\Delta \xrightarrow{\delta} \mathcal{O}_X(-D) \boxtimes \omega_X(D)[n],$$

which amounts to checking that ε is in the image of precomposition by ρ^*

$$\operatorname{Hom}(\mathcal{O}_{X\times X}, \mathcal{O}_X(-D) \boxtimes \omega_X(D)[n]) \leftarrow \operatorname{Hom}(\mathcal{O}_\Delta, \mathcal{O}_X(-D) \boxtimes \omega_X(D)[n]).$$

This is the same sort of story as before. By Serre duality, ρ^* corresponds to a morphism

$$\operatorname{Hom}(\mathcal{O}_X(-D) \boxtimes \omega_X(D), \omega_{X\times X}[n])^\vee \leftarrow \operatorname{Hom}(\mathcal{O}_X(-D) \boxtimes \omega_X(D), \mathcal{O}_\Delta \otimes \omega_{X\times X}[n])^\vee,$$

which is dual to restriction $\omega_{X\times X} \to \mathcal{O}_\Delta \otimes \omega_{X\times X}$ along the diagonal. Making further identifications, this is a morphism

$$\begin{aligned} H^n(X\times X, \omega_X(D) \boxtimes \mathcal{O}_X(-D))^\vee \leftarrow & H^n(X\times X, \mathcal{O}_\Delta \otimes (\omega_X(D), \mathcal{O}_X(-D)))^\vee \\ & \cong H^n(X, \omega_X(D) \otimes \mathcal{O}_X(-D))^\vee, \end{aligned}$$

and by the Kunneth formula and Kodaira vanishing, this left hand term is just

$$(H^0(X, \omega_X(D)) \otimes H^n(X, \mathcal{O}_X(-D)))^\vee.$$

Then, one can inspect that the morphism

$$(H^0(X, \omega_X(D)) \otimes H^n(X, \mathcal{O}_X(-D)))^\vee \leftarrow H^n(X, \omega_X(D) \otimes \mathcal{O}_X(-D))^\vee \tag{2}$$

is dual to the usual cup product of cocycle classes/composition in the derived category. Now, we try to reach ε. We pick an isomorphism $H^n(X, \omega_X) \cong k$, a basis $x_1, \ldots, x_\ell$ of

$H^0(X, \omega_X(D))$, and the Serre dual basis $e_1, \ldots, e_\ell$ for $H^n(X, \mathcal{O}_X(-D))$. Then, ε is identified with

$$\sum e_i \otimes x_i,$$

(notice the x and e are swapped, because it is a functional). At this point, it is quick to see this ε is indeed reached, since dual to (2) is, in our basis,

$$\begin{aligned} H^0(X, \omega_X(D)) \otimes H^n(X, \mathcal{O}_X(-D)) &\longrightarrow k \\ x_i \otimes e_j &\longmapsto \delta_{ij}, \end{aligned}$$

and this functional itself *is* $\varepsilon = \sum e_i \otimes x_i$. That is, 1_k is a functional on $H^n(X, \omega_X)$, and pulling back the Yoneda cup product/composition, we send 1_k in (2) to ε. Thus, we get our desired factorization.

Now that we have the factorization, denote by C_Δ the cone over

$$\mathcal{O}_\Delta \to \mathcal{O}_X(-D) \boxtimes \omega_X(D)[n],$$

so that the octahedral axiom applied to our composition gives the triangle

$$\to \mathcal{I}_\Delta \to C_X \to C_\Delta \to .$$

Similar to before, we try to understand C_Δ by applying derived dual $(-)^\vee = R\mathcal{H}om(-, \mathcal{O}_{X\times X})$ to the triangle defining C_Δ. We apply the functor term-by-term. Applying Grothendieck duality for $i : X \to \Delta \hookrightarrow X \times X$ the inclusion of the diagonal, we have

$$\begin{aligned} R\mathcal{H}om_{X\times X}(\mathcal{O}_\Delta, \mathcal{O}_{X\times X}) &\cong R\mathcal{H}om_{X\times X}(Ri_*\mathcal{O}_X, \mathcal{O}_{X\times X}) \\ &\cong Ri_* R\mathcal{H}om_X(\mathcal{O}_X, i^*\mathcal{O}_{X\times X} \otimes \omega_X \otimes i^*(\omega_{X\times X}^{-1})[-n]) \\ &= Ri_* R\mathcal{H}om_X(\mathcal{O}_X, \omega_X^{-1}[-n]) \\ &= i_*\omega_X^{-1}[-n]. \end{aligned}$$

Next,

$$R\mathcal{H}om_{X\times X}(\mathcal{O}_X(-D) \boxtimes \omega_X(D)[n], \mathcal{O}_{X\times X}) \cong \mathcal{O}_X(D) \boxtimes \omega_X^{-1}(-D)[-n],$$

so we have the triangle

$$\leftarrow C_\Delta^\vee \leftarrow i_*\omega_X^{-1}[-n] \leftarrow \mathcal{O}_X(D) \boxtimes \omega_X^{-1}(-D)[-n] \leftarrow,$$

which has the advantage that the morphism of the right two terms is a (shift of a) nonzero morphism

$$\mathcal{O}_X(D) \boxtimes \omega_X^{-1}(-D) \xrightarrow{\varphi} i_*\omega_X$$

of sheaves in

$$\begin{aligned} \mathrm{Hom}(\mathcal{O}(D) \boxtimes \omega_X^{-1}(-D), i_*\omega_X^{-1}) &= \mathrm{Hom}(i^*(\mathcal{O}(D) \boxtimes \omega_X^{-1}(-D)), \omega_X^{-1}) \\ &= \mathrm{Hom}(\omega_X^{-1}, \omega_X^{-1}) \end{aligned}$$

so it is unique up to scaling. In fact, we can recognize the target of φ as

$$i_*i^*(\mathcal{O}(D) \boxtimes \omega_X^{-1}(-D)) \cong i_*\omega_X^{-1},$$

so (up to scaling) our morphism has to be the unit map of the $i^* \dashv i_*$ adjunction, which can be computed, by the projection formula, as

$$(\mathcal{O}_{X\times X} \to i_* i^*(\mathcal{O}_{X\times X})) \otimes (\mathcal{O}(D) \boxtimes \omega_X^{-1}(-D)),$$

which makes it clear that φ is a surjective map with kernel

$$\mathcal{I}_\Delta \otimes (\mathcal{O}_X(D) \boxtimes \omega_X^{-1}(-D)),$$

which implies

$$C_\Delta^\vee \cong \mathcal{I}_\Delta \otimes (\mathcal{O}_X(D) \boxtimes \omega_X^{-1}(-D))[1-n],$$

so

$$C_\Delta \cong \mathcal{I}_\Delta^\vee \otimes (\mathcal{O}_X(-D) \boxtimes \omega_X(D))[n-1].$$

Therefore, our triangle describing the family C_X over X becomes

$$\to \mathcal{I}_\Delta \to C_X \to \mathcal{I}_\Delta^\vee \otimes (\mathcal{O}_X(-D) \boxtimes \omega_X(D))[n-1] \to .$$

2.2 Elementary Modifications

We take a moment to talk about the general theory of elementary modifications of complexes in the derived category. Let X smooth projective, $i : D \hookrightarrow X$ a smooth (effective) Cartier divisor, and $C \in D^b(X)$. We want to modify C along the divisor D.

We modify C according to a "quotient" of C along the divisor. That is, we take as input a distinguished triangle

$$\to K \to Li^*C \xrightarrow{f} Q \to$$

in $D^b(D)$. Then, we have the composition

$$C \to i_* Li^* C \xrightarrow{i_* f} i_* Q$$

of the unit of the $Li_* \dashv i_*$ adjunction with $Ri_* f$. Denote B the cone over this composition, which is what we will call our elementary modification of C.

In other words, B is defined by the following distinguished triangle

$$\to B \to C \to i_* Q \to .$$

This elementary modification is "reversible" in the following sense:

Proposition 2.2. The elementary modification of the complex C on X along $i : D \hookrightarrow X$ by $Li^*C \to Q$ fits into a distinguished triangle

$$\to Q \otimes \mathcal{O}_D(-D) \to Li^*B \to K \to,$$

and the elementary modification of B along D by this $Li^*B \to K$ recovers $C(-D)$.

Remark 2.3. We first say a word about the proof in the classical elementary modification setting, where C, Q are vector bundles, and f is a surjection. The proof is simply to consider the short exact sequence defining B (now as a kernel), take the long exact sequence in $\mathcal{T}or^{\mathcal{O}_X}(-, \mathcal{O}_D)$ to pullback to get

$$0 \to \mathcal{T}or_1^{\mathcal{O}_X}(Q, \mathcal{O}_D) \to i^*B \to i^*C \to Q \to 0$$

(by flatness of i^*C). Then, you focus on the first three terms instead of the last (in the derived category, this will be "rotating the triangle"). Last, replace i^*C with the image of i^*B, which is exactly K, and compute $\mathcal{T}or_1^{\mathcal{O}_X}(Q, \mathcal{O}_D) \cong Q(-D)$.

The proof will be different in the derived setting, because we will instead be using derived pullback Li^*, which already incorporates all $\mathcal{T}or^{\mathcal{O}_X}(-, \mathcal{O}_D)$ and is also exact.

Proof. Pull back the distinguished triangle defining B to get

$$\to Li^*B \to Li^*C \to Li^*i_*Q \to .$$

Already this is different from the classical setting, because Li^*i_*Q is *not* Q, we have twisted Q with normal data of D. We do however have the counit map $\varepsilon_Q : Li^*i_*Q \to Q$ for the $Li^* \dashv i_*$ adjunction, and applying the octahedral axiom to the composition

$$Li^*C \xrightarrow{q} Li^*i_*Q \xrightarrow{\varepsilon_Q} Q,$$

which is none other than starting map $Li^*C \xrightarrow{f} Q$, we get the triangle

$$\to C_q \to C_f \to C_{\varepsilon_Q} \to,$$

which simplifies (in particular using 3.3 for C_{ε_Q}) to the triangle

$$\to Li^*B \to K \to Q \otimes \mathcal{O}_D(-D)[1] \to,$$

which rotates to our desired triangle

$$\to Q \otimes \mathcal{O}_D(-D) \to Li^*B \to K \to .$$

Then, one can check that the cone over the composition

$$B \to i_*Li^*B \to i_*K$$

recovers $C(-D)$, since the octahedral axiom places such a cone it in a distinguished triangle

$$\to B \otimes \mathcal{O}_X(-D) \to (-) \to i_*Q \otimes \mathcal{O}_X(-D) \to,$$

so one only needs to check that the morphism $i_*Q \otimes \mathcal{O}_X(-D) \to B \otimes \mathcal{O}_X(-D)[1]$ is correct. probably follows from all of the commuting diagrams in the octahedral axiom.

□

3 Appendix

3.1 Derived Self-Intersection

Let X be a smooth projective variety, and $i : Z \hookrightarrow X$ be a codimension r smooth closed subscheme cut out by a regular section s of a vector bundle E on X.

We will compute the **derived self-intersection**, $Li^* Ri_* \mathcal{O}_Z$. Note that i is an affine map, so $Ri_* = i_*$.

Proposition 3.1. $Li^* i_* \mathcal{O}_Z = \bigoplus_{i=0}^r \wedge^i E^\vee|_D[i]$

Proof. The section $s \in \Gamma(X, E)$ exactly gives a surjection

$$\wedge^1 E^\vee = E^\vee \xrightarrow{s} \mathcal{I}_Z \hookrightarrow \mathcal{O}_X,$$

which starts a Koszul complex $\mathrm{Kos}(s)$ quasi-isomorphic to $i_* \mathcal{O}_Z$,

$$\mathrm{Kos}(s) = \left(0 \to \wedge^r E^\vee \to \cdots \to \wedge^1 E^\vee \to \wedge^0 E^\vee \to 0\right).$$

Now, $\mathrm{Kos}(s)$ consists entirely of locally free sheaves, so we can compute Li^* just by usual pullback i^*. Furthermore, on trivialization of E, where s in coordinates is $(f_1, \dots, f_r)$, the differentials consist of matrices of the f_i, but these become 0 on Z. Thus, $i^* \mathrm{Kos}(s)$ is just

$$\bigoplus_{i=0}^r \wedge^i E^\vee|_D[i],$$

which computes $Li^* i_* \mathcal{O}_Z$. □

Proposition 3.2. Let $K \in D^b(Z)$, then

$$Li^* i_* K = \bigoplus_{i=0}^r \wedge^i E^\vee|_D \otimes K[i].$$

Proof. Recall that, by definition,

$$i^*(-) \cong i^{-1}(-) \otimes^L_{i^{-1}\mathcal{O}_X} \mathcal{O}_Z,$$

so "deriving both sides", we get

$$Li^*(-) \cong i^{-1}(-) \otimes^L_{i^{-1}\mathcal{O}_X} \mathcal{O}_Z.$$

The advantage of this formulation is that i^{-1} is an exact functor and left inverse to i_* in the derived category.

Note, we are slightly abusing notation to write i_* to refer to both functors

$$D^b(Z) \to D^b(X), \qquad D^b(i^{-1}\mathcal{O}_X\text{-Mod}) \to D^b(\mathcal{O}_X\text{-Mod}),$$

(since both functors are computed in the same way on the underlying sheaf of abelian groups). These two are related by $i_*(-)$ in the first sense being the same as $i_*((-)|_{i^{-1}\mathcal{O}_X})$ for the second meaning of i_*, but this is essentially just type-checking.

So, we have

$$Li^*i_*(K) \cong i^{-1}i_*(K) \otimes^L_{i^{-1}\mathcal{O}_X} \mathcal{O}_Z.$$

Now recall that for s the section of E cutting out Z, $i_*\mathcal{O}_Z$ as a $\mathcal{O}_X$-module is quasi-isomorphic to $\operatorname{Kos}(s)$, so that

$$\mathcal{O}_Z \cong i^{-1}i_*\mathcal{O}_Z \cong i^{-1}\operatorname{Kos}(s),$$

where $i^{-1}\operatorname{Kos}(s)$ is a complex of flat $i^{-1}\mathcal{O}_X$-modules. Now, we get

$$\begin{aligned} Li^*i_*(K) &\cong K \otimes_{i^{-1}\mathcal{O}_X} i^{-1}\operatorname{Kos}(s) \\ &\cong K \otimes_{\mathcal{O}_D} \mathcal{O}_D \otimes_{i^{-1}\mathcal{O}_X} i^{-1}\operatorname{Kos}(s) \\ &\cong K \otimes_{\mathcal{O}_D} i^*\operatorname{Kos}(s) \\ &\cong \bigoplus_{i=0}^{r} \wedge^i E^\vee|_D \otimes_{\mathcal{O}_D} K[i]. \end{aligned}$$

□

Corollary 3.3. The cone over the counit map $\varepsilon_K : Li^*i_*K \to K$ of the $Li^* \dashv i_*$ adjunction is (quasi-)isomorphic to

$$\bigoplus_{i=1}^{r} \wedge^i E^\vee|_D \otimes K[i].$$

Proof. Appropriately expressed, the adjunction $Li^* \dashv i_*$ lifts to the category of chain complexes as the composition of the $i^{-1} \dashv i_*$ adjunction and a certain tensor-hom adjunction, as in

$$\begin{aligned} i^{-1}(-) \otimes_{i^{-1}\mathcal{O}_X} i^{-1}\operatorname{Kos}(s) \quad &\dashv \quad i_*\mathcal{H}om_{i^{-1}\mathcal{O}_X}(i^{-1}\operatorname{Kos}(s), -) \\ &\cong \mathcal{H}om_{\mathcal{O}_X}(\operatorname{Kos}(s), i_*(-)), \end{aligned}$$

where nothing is derived, which gives us a chain isomorphism

$$\operatorname{Hom}(i^{-1}(-) \otimes_{i^{-1}\mathcal{O}_X} i^{-1}\operatorname{Kos}(s), -) \cong \operatorname{Hom}(-, \mathcal{H}om(\operatorname{Kos}(s), i_*(-)).$$

Now, the counit is the image of the identity chain map under this isomorphism in the following sense:

$$\begin{aligned} \operatorname{Hom}(\mathcal{H}om(\operatorname{Kos}(s), i_*(K)), \mathcal{H}om(\operatorname{Kos}(s), i_*(K))) &\xrightarrow{\cong} \operatorname{Hom}(i^{-1}(\mathcal{H}om(\operatorname{Kos}(s), i_*(K))) \otimes_{i^{-1}\mathcal{O}_X} i^{-1}\operatorname{Kos}(s), K) \\ &\cong \operatorname{Hom}(\mathcal{H}om(i^{-1}\operatorname{Kos}(s), K) \otimes_{i^{-1}\mathcal{O}_X} i^{-1}\operatorname{Kos}(s), K) \\ 1 &\mapsto \varepsilon_K, \end{aligned}$$

so ε_K is essentially evaluation. As in the proof of the previous proposition, we can "throw in" extra $\mathcal{O}_D$ tensor factors using the fact that K is a $\mathcal{O}_D$ module, so that in fact ε_K becomes a more transparent evaluation in

$$\mathrm{Hom}(\mathcal{H}om(i^* \operatorname{Kos}(s), K) \otimes_{\mathcal{O}_D} i^* \operatorname{Kos}(s), K).$$

Again writing $i^* \operatorname{Kos}(s) \cong \bigoplus_{i=0}^r \wedge^i E^\vee|_D[i]$, we see that ε_K is the evaluation

$$\mathrm{Hom}(\bigoplus_{i=0}^{r})$$

no i'm doing something bs

$\square$